Installing Helm

Any Kubernetes installation tool like Vanilla Kubernetes, Minikube, Kind, Red Hat OpenShift, EKS, and Azure KS can be used for installing Helm. Installing Kubernetes cluster is beyond the scope of this article.

You can install Helm on Windows, Linux and macOS.

On Windows, you can use Chocolatey:

```
> choco install kubernetes-helm
```

On macOS, you can use Homebrew:

```
$ brew install helm
```

On Linux, you can use package managers like yum, dnf, snap or apt:

```
$ apt install helm
```

You can also use direct download links for Helm GitHub repository (*https://github.com/helm/helm/releases):*

```
$ wget https://get.helm.sh/helm-v3.2.1-
linux-amd64.ttar.gz
$ tar xvzf helm-v3.2.1-linux-amd64.tar.
gz
$ mv linux-amd64/helm/usr/local/bin/
helm
```

You can verify the Helm installation by running:

```
$ helm version
version.BuildInfo{Version:"v3.2.1", Gi
tCommit:"fe51cd1e31e6a202cba7dead9552
a6d418ded79a", GitTreeState:"clean",
GoVersion:"go1.13.10"}
```

Helm environment variables

Helm allows configuring the following variables:

```
HELM_BIN
HELM_NAMESPACE
HELM_DEBUG
HELM_PLUGINS
HELM_KUBEAPISERVER
HELM_REGISTRY_CONFIG
HELM_KUBECONTEXT
HELM_REPOSITORY_CACHE
HELM_KUBETOKEN
HELM_REPOSITORY_CONFIG
```

You can read more about these configurations in the Helm Official documentation at *https://helm.sh/docs/.*

You can configure Helm environment variables by exporting variables using the *export* command:

```
$ export HELM_DEBUG=True
$ helm env
```

Basic commands

Managing upstream chart repositories: Helm downloads charts from remote or local repositories. You can manage repositories' operations using commands such as:

- *add:* To add a repository
- *list:* To list chart repositories
- *remove*: To remove chart repository
- *update:* To update available chart information

Adding chart repository: Chart repositories are required to install charts. You can add a chart repository by using the following command:

```
$ helm repo add <REPO_NAME> <REPO_
URL>
```

Listing chart repositories: The following command can be used to list all installed chart repositories:

```
$ helm repo list
```

Updating chart repositories: You can check chart updates from the given repositories by issuing the following command:

```
$ helm repo update
```

Removing chart repository: You can remove a repository by using the following command:

```
$ helm repo remove <REPO_NAME>
```

```
 $ helm search repo wordpress
NAME                 CHART VERSION   APP VERSION   DESCRIPTION

bitnami/wordpress    9.5.1           5.5.0         Web publishing platform for building bl
ogs and ...
stable/wordpress     9.0.3           5.3.2         DEPRECATED Web publishing platform for
building...
```

Figure 1: Helm search

```
 $ helm install wordpress bitnami/wordpress --values=values.yml --namespace=default
NAME: wordpress
LAST DEPLOYED: Mon Aug 31 22:32:13 2020
NAMESPACE: default
STATUS: deployed
REVISION: 1
NOTES:
** Please be patient while the chart is being deployed **

Your WordPress site can be accessed through the following DNS name from within your cluster:

    wordpress.default.svc.cluster.local (port 80)

To access your WordPress site from outside the cluster follow the steps below:

1. Get the WordPress URL by running these commands:

   export NODE_PORT=$(kubectl get --namespace default -o jsonpath="{.spec.ports[0].nodePort}" s
ervices wordpress)
   export NODE_IP=$(kubectl get nodes --namespace default -o jsonpath="{.items[0].status.addres
ses[0].address}")
   echo "WordPress URL: http://$NODE_IP:$NODE_PORT/"
   echo "WordPress Admin URL: http://$NODE_IP:$NODE_PORT/admin"
```

Figure 2: Helm install